Neil (Hsuan Chih) Chen

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Education

New York University May 2026
Master's in **Data Science** Cumulative GPA: 3.8/4.0

University of Minnesota-Twin Cities May 2023
Bachelor of Arts in **Mathematics** Cumulative GPA: 3.6/4.0
Minor in **Statistics**

Research Experience

University of Pennsylvania

Philadelphia, PA

Researcher, Han's AI Lab June 2026 – Present
Supervisor: Dr. Tianyu Han

JEV-Agent: Explainable and Efficient Clinical Decisions from EHRs
September 2026 – Present
Manuscript in preparation

- Developed an EHR-based clinical workflow using JEV, a decision model for structured options, to select tools, acquire evidence, and update differential diagnoses.
- Benchmarked diagnostic accuracy, inference cost, latency, and consistency against standalone LLMs and clinical agents.
- Made clinical decisions traceable through action probabilities, evolving differential diagnoses, and evidence-linked work-up logs.
- Achieved 86.5% mean diagnostic accuracy, outperforming evaluated agent baselines on pulmonary embolism and meningitis.
- Processed 546 cases for $1.31 versus $20.50–$79.70 for single-run agent baselines, with median runtimes of 2.2 versus 89.6–99.6 seconds per case.

Med-JEV: Structured Decision Learning from Medical Images
September 2026 – Present
Work in progress

- Developing a teacher–student framework that distills radiology reports into question-specific labels and probabilities for medical image interpretation.
- Designed and implemented a pipeline to generate, sample, and validate questions from existing datasets using LLMs and JEV.
- Adapted a MedGemma student with a shared scoring head to learn binary, categorical, and ordinal decisions from JEV supervision.

RL Attacker: Indirect Prompt Injection in Medical Agents
June 2026 – Present
Work in progress

- Developed a controlled framework to evaluate MEDDxAgent's vulnerability to indirect prompt injection in retrieved external content, using frozen retrieval and paired clean and attacked runs.
- Trained a Llama-3.1-8B attacker with GRPO and LoRA to generate injected instructions that promote a targeted incorrect diagnosis.

- Raised targeted attack success (rank-1 flips) from 31.9% with manual injection to 46.6% with RL (+14.7 percentage points) on 251 eligible held-out cases; single-seed evaluation with oracle-assisted targets.

New York University New York, NY

Research Assistant, Shen's Lab April 2025 – June 2026
Supervisor: Dr. Yiqiu Shen

DeepMS: Multimodal AI for Multiple Sclerosis Diagnosis
April 2025 – August 2025
Under revision for JAMA Neurology

- Integrated age and sex into DeepMS, a multimodal MRI framework for multiple sclerosis diagnosis.
- Compared demographic–imaging fusion strategies to assess the contribution of patient metadata to diagnostic performance.

Multimodal and Longitudinal Breast Cancer Detection
September 2025 – Present
Manuscript in preparation

- Evaluated a multimodal transformer integrating current and prior 2D and 3D mammography, ultrasound, and clinical data for breast cancer detection.
- Curated public datasets for external validation and packaged the inference pipeline in Docker for private institutions to evaluate on their own data.
- Internal evaluation achieved an AUROC of 0.933 (95% CI: 0.921–0.944); external AUROCs ranged from 0.831 to 0.958 across institutional cohorts and public benchmarks.
- Developed attention visualizations to examine image regions associated with model predictions across modalities.

Multimodal AI for Acute Respiratory Failure Prediction
September 2025 – January 2026

- Developed a late-fusion framework combining CheXFound imaging predictions and XGBoost clinical predictions to classify acute respiratory failure etiologies and predict mortality.
- Achieved a 28-day mortality AUROC of 0.835 with fusion, compared with 0.802 for imaging alone and 0.702 for clinical data alone.

Publications and Preprints

- Ma, J., Stepanov, V., Rui, W., **Chen, H.-C.**, et al. (2026). Diagnosis of Multiple Sclerosis Using Multimodal Deep Learning Integrating Lesion and Normal-Appearing White Matter: A Retrospective Study with International Multicentre External Validation. *medRxiv* [Preprint]. Under revision for *JAMA Neurology*.

Manuscripts in Preparation

- JEV-Agent: Explainable and Efficient Clinical Decisions from EHRs.
- Multimodal and Longitudinal Breast Cancer Detection Using Imaging and Clinical Data.

Work in Progress

- Med-JEV: Structured Decision Learning from Medical Images.
- RL Attacker: Indirect Prompt Injection in Medical Agents.